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A-A-59503D
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SUPERSEDING
A-A-59503C
3 March 2011

# COMMERCIAL ITEM DESCRIPTION

# NITROGEN, TECHNICAL

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers two types of nitrogen: gaseous and liquid. Each type of nitrogen can be obtained in three different technical grades based primarily on nitrogen and oxygen content. Liquid nitrogen is used as a cooling agent for low temperature and cryogenic processes to shield temperature-sensitive materials and equipment from the effects of heat. Gaseous nitrogen is used to purge or pressurize systems or provide inert atmospheres. The list of intended uses includes, but is not limited to: pressurizing fuel tanks, hydraulic system accumulators, aircraft struts, rocket engine propellant systems, and carbon dioxide cylinders; purging aircraft oxygen converters; and purging and calibrating instruments. Since some systems are sensitive to oil contamination, gaseous nitrogen is divided into two classes: oil free and oil tolerant. Class 1, oil free nitrogen, is used for applications that cannot tolerate hydrocarbons, such as the purging of oxygen equipment. Class 2, oil tolerant nitrogen, is used for pressurizing oil-containing systems such as aircraft struts.

2. CLASSIFICATION. The nitrogen shall be classified by the following types, grades, and classes:

2.1 Type. The technical nitrogen shall be one of the following types, as specified in the acquisition order (see 7.4):

Type I - Gaseous
Type II - Liquid

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AMSC N/A FSC 6830

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2.2 Grade. Nitrogen of grades UHP and A are equivalent to quality verification levels (QVLs) presented in Compressed Gas Association (CGA) G-10.1, "Commodity Specification for Nitrogen". The technical nitrogen shall be one of the following grades, as specified in the acquisition order (see 7.4):

Grade UHP - 99.999 percent nitrogen (equivalent to QVL Q)
Grade A - 99.998 percent nitrogen (equivalent to QVL L)
Grade B - 99.50 percent nitrogen

2.3 Class. The terms "oil free" and "oil tolerant", indicated by Class 1 and Class 2, replaced the older terms "water pumped" and "oil pumped", respectively. The older terminology referenced the type of compressors, water and soap lubricated, or oil lubricated, used in charging nitrogen gas into cylinders. Due to advancements in compressor technology and other systems being used, the newer terms now refer to the type of application. When applied to nitrogen, the term "oil free" shall be used to identify applications that cannot tolerate hydrocarbon contamination. The term "oil tolerant", when applied to nitrogen, indicates that the gas can be used in applications that can tolerate or contain hydrocarbon material. This terminology applies only to gaseous nitrogen (see 7.5). The technical nitrogen shall be one of the following classes, as specified in the acquisition order (see 7.4):

Class 1 - Oil free
Class 2 - Oil tolerant (Type I only)

## 3. SALIENT CHARACTERISTICS

3.1 Material. The material shall conform to the requirements as specified in Table I. The limiting characteristics shall be determined by one of the methods described in CGA G-10.1.

TABLE I. Requirements. [1]

| Limiting characteristics | Grade UHP [2] | Grade A [2] | Grade B [2] |
|---|---|---|---|
| CGA G-10.1 Quality Verification Level | QVL Q | QVL L | n/a |
| Nitrogen, minimum (percent by volume) [3] | 99.999 | 99.998 | 99.50 |
| Argon, neon, helium | 5 | - | - |
| Oxygen | 1 | 10 | 5,000 |
| Water vapor | 2 | 4 | 26 |
| Total hydrocarbon content (as methane) | 1 | 20 | 50 |

[1] Additional information on the properties and handling of nitrogen may be found in CGA P-9, "The Inert Gases: Argon, Nitrogen, and Helium".
[2] Unless otherwise indicated, quantities given are the maximum in parts per million by volume (ppm (v/v)).
[3] Unless otherwise indicated, the percent by volume of nitrogen for Type I may include up to a total of 1.5% by volume of argon, neon, and helium.

3.2 Odor. The nitrogen shall have no odor.



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3.3 Particulate matter. The material shall contain no solid particles whose maximum dimension is greater than 50 micrometers (µm) for Type I nitrogen or 30 µm for Type II nitrogen, when specified in the acquisition order (see 7.4). This shall be determined by any acceptable commercial method practiced by the manufacturers of nitrogen. Because the major source of contamination is from cylinders and other containment or transfer systems, control of particulate matter in the supplier's and user's dispensing equipment can be assured by the installation of a 10-µm (nominal) or better filter in the service line.

## 4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Safety data sheet (SDS). Manufacturers shall prepare and submit a SDS in accordance with FED-STD-313, "Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities", and meeting the requirements of 29 Code of Federal Regulations (CFR) §1910.1200, "Occupational Safety and Health Standards: Hazard Communication".

4.3 Hazard warning label. Occupational Safety and Health Administration hazard warning labels shall be furnished in accordance with 29 CFR 1910.1200.

## 5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The product provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Market acceptability. The products offered must have been previously sold either to the government or on the commercial market.

5.2.1 Market acceptability criterion. The company must be able to show data from tests or process monitoring that demonstrates the ability of the product to meet the salient characteristics of technical grade nitrogen as specified herein.

## 6. PACKAGING

6.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the acquisition order (see 7.4).

6.2 Packaging and special markings. All government-owned cylinders that have been provided for filling shall be reconditioned as necessary in accordance with MIL-STD-1411. Department of Defense cylinders shall conform to RR-C-901, be equipped with valves conforming to



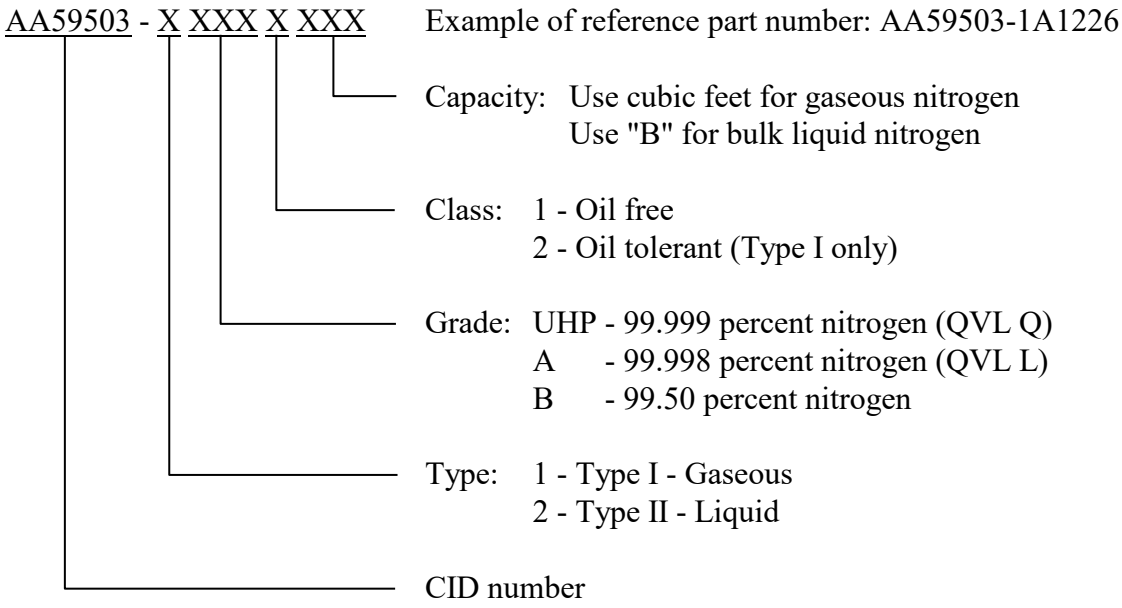
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A-A-59860, and be color-coded in accordance with MIL-STD-101. The capacity of nitrogen shall be specified in cubic feet for gaseous nitrogen and designated by the letter "B" for bulk liquid nitrogen or as specified in the acquisition order (see 7.4).

6.3 Palletization. The palletization of material shall be as specified in the acquisition order (see 7.4). MIL-STD-147 may be used for palletization.

7. NOTES

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.




7.2 Sources of documents.

7.2.1 Electronic copies of federal specifications and standards can be obtained from https://quicksearch.dla.mil.

7.2.2 Electronic copies CFR documents can be obtained from https://gov.ecfr.io/cgi-bin/ECFR.

7.2.3 Electronic copies of FAR documents may be obtained from https://www.acquisition.gov/.

7.2.4 Electronic copies of CGA standards may be obtained from http://www.cganet.com/.

7.3 International standardization agreements (ISAs). Certain provisions of this CID (section 3) are subject to ISA NATO STANAG 3624, "Nitrogen and Replenishment Equipment Characteristics". When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change



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the agreement or make other appropriate accommodations. Copies of this ISA are available online at https://assist.dla.mil.

7.4 Ordering data. The acquisition order should specify the following information:

a. CID document number, revision, and CID PIN.
b. Type of technical nitrogen (see 2.1).
c. Grade of technical nitrogen (see 2.2).
d. Class of technical nitrogen (see 2.3).
e. Particulate matter testing, if required (see 3.3).
f. Preservation, packing, and marking requirements (see 6.1).
g. Capacity, in cubic feet for gaseous nitrogen, or "B" for bulk liquid nitrogen (see 6.2).
h. Palletization requirements (see 6.3).

7.5 Intended use. Oil tolerant nitrogen gas, Class 2, shall only be used with oil-containing systems or systems that can tolerate hydrocarbon contamination, like pressurizing aircraft struts. It is not to be used to purge or pressurize oxygen or air for human respiration systems or for any vessel piping, valves, gauges, etc. that is for oxygen service. Contamination of an oxygen system with an oil tolerant gas could result in a fire or explosion with loss of life or loss of a complete weapons system. This CID is not suitable for procurement of nitrogen for use in reactor plant, steam plant, and shipyard applications. For these applications see A-A-59155.

7.6 Valve outlet connections.

7.6.1 Cautionary note. The valve outlet connection for oil tolerant nitrogen should be different from oil free valve outlet connections to prevent the erroneous use of oil tolerant nitrogen cylinders for oil free applications. A reverse flow typically occurs during an oil tolerant operation (pressurizing oil-containing systems, e.g. aircraft struts). As a result, cylinders are contaminated with oil and cannot be used for oil free applications.

7.6.2 Valve connections. The applicable standard for valve connections is CGA V-1, "Standard for Compressed Gas Cylinder Valve Outlet and Inlet Connections". Note that valve connection 590 is a limited standard for nitrogen per CGA V-1. The connections listed in table II apply.

TABLE II. Valve connection numbers.

| Pressure (psig) | Up to 3,000 | 3,001 - 4700 |
|---|---|---|
| Class 1 (oil free) | 580 | 680 |
| Class 2 (oil tolerant) | 580 or 590 | 621 |

7.7 Subject term (key word) listing.

Aviator's breathing oxygen
Calibrate



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Compressed gas
Cooling agent
Cryogenic
Cylinder
Human respiration
Inert
Oil free
Oil tolerant
Pressurize
Purge
Struts, aircraft
Valves

7.8 Changes from previous issues. Marginal notations are used in this revision to identify changes with respect to the previous issue.

| MILITARY INTERESTS: | CIVIL AGENCY COORDINATING ACTIVITY: |
|---|---|
| Custodians:<br>Army - AV<br>Navy - SH<br>Air Force - 68<br>DLA - GS | GSA - FAS |
| Review Activities:<br>Army - AR, MD1<br>Navy - AS, MC, OS, YD<br>Other - DS | Preparing Activity:<br>DLA - GS3<br><br>(Project 6830-2019-001) |

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